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# Drive Test Throughput Anomaly Detection Report

## LTE Downlink Throughput, Anomaly Detection, and RCA Pipeline

From Cleaned Drive-Test Data to RCA-Ready Outputs

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## 1 Executive Summary

The drive-test project will be divided into three connected parts. The first part is the anomaly detection module, which detects abnormal LTE downlink throughput behavior. The second part is the root cause analysis (RCA) module, which explains why the anomaly happened. The third part is the LLM module, which converts the anomaly evidence and RCA evidence into a clear structured explanation.

### Recommended First Scope

The first version should focus on LTE downlink throughput anomalies using the cleaned drive-test data. The model should not try to solve all technologies and all services at once. The best starting point is LTE download sessions, then the same pipeline can be extended later with handover, RACH, carrier aggregation, HTTP-layer throughput, and WCDMA/CSFB features.

The main target is not only to detect low throughput. The more useful objective is to classify throughput degradation into clear anomaly types:

- **Radio-limited degradation:** low throughput caused by poor RSRP, RSRQ, SINR, or high BLER.
- **Unexpected underperformance:** actual throughput is much lower than expected although the radio conditions look good.
- **Carrier aggregation underperformance:** carrier aggregation is active or expected, but throughput remains poor.
- **Mobility or event-related degradation:** throughput degradation near handover, RACH, CA activation/deactivation, or other network events.
- **Sudden throughput drop:** throughput drops sharply inside a session and the pipeline checks whether radio degraded at the same time.

## 2 Project Direction and Data Role

## 2.1 System Split

| Part              | Goal                                                                        | Main Output                                                                                                    |
|-------------------|-----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Anomaly detection | Detect throughput degradation and classify the anomaly type.                | RCA-ready anomaly table with timestamp, location, throughput gap, severity, and evidence flags.                |
| RCA               | Explain the most likely technical cause behind each anomaly.                | Cause labels such as poor coverage, poor quality, high BLER, low scheduling, CA issue, or event-related issue. |
| LLM layer         | Transform anomaly and RCA evidence into a clean human-readable explanation. | Structured output for reports, dashboards, and engineer-facing summaries.                                      |

**Table 1:** High-level project split.

## 2.2 Why LTE Throughput First?

The cleaned data shows that LTE is the strongest branch for machine learning. The NR5G measurement columns are not usable in the current drive-test file, and the planning file is geographically mismatched with the drive-test route. Therefore, the first practical project should focus on LTE/WCDMA drive-test performance, with LTE downlink throughput anomaly detection as the first module.

The LTE-DL part is suitable because it contains enough samples, useful radio measurements, scheduling/PHY features, carrier aggregation information, GPS route data, and event information that can later be converted into RCA evidence.

## 3 Cleaned Data Preparation

### 3.1 Main Cleaned Dataset

The main row-level cleaned file is the cleaned base drive-test table. It keeps all samples after removing highly useless columns, while preserving important time, location, LTE, WCDMA, activity, and event fields.

For anomaly detection V1, the full cleaned table should not be used blindly. The first modeling table should be filtered to LTE downlink sessions only.

### Recommended V1 Filter

Use rows where LTE is active, LTE throughput exists, and the test type is a download test. MO-call and WCDMA-heavy rows should be kept for later RCA modules, but should not be mixed into the first LTE throughput model.

## 3.2 Initial Row Filter

The first LTE-DL modeling table should be created using logic similar to the following:

```
df_lte = df[
    (df["has_lte_measurement"] == True) &
    (df["has_lte_throughput"] == True) &
    (df["id_test_type"].str.contains("DL", na=False))
].copy()

# Canonical target
df_lte["lte_dl_throughput"] = df_lte["LTE.LTE_Data_KPI.Agg_Throughput_DL"]

# Keep valid target rows only
df_lte = df_lte[df_lte["lte_dl_throughput"].notna()]
df_lte = df_lte[df_lte["lte_dl_throughput"] >= 0]
```

## 3.3 Canonical Column Strategy

Because drive-test exports usually have long and inconsistent column names, the anomaly notebook should create short canonical names. This makes modeling, plotting, and RCA easier.

| Canonical Name    | Meaning                              | RCA Use                                                     |
|-------------------|--------------------------------------|-------------------------------------------------------------|
| lte_dl_throughput | Main LTE downlink throughput target. | Detect throughput degradation and calculate throughput gap. |
| lte_rsrp          | LTE serving or primary-cell RSRP.    | Coverage-limited RCA.                                       |
| lte_rsrq          | LTE serving or primary-cell RSRQ.    | Quality/interference/load RCA.                              |
| lte_sinr          | LTE serving or primary-cell SINR.    | Radio quality and expected throughput.                      |
| lte_cqi           | Channel quality indicator.           | Link adaptation and expected MCS/throughput.                |
| lte_mcs           | Modulation and coding scheme.        | Indicates selected coding/modulation level.                 |
| lte_bler          | Block error rate.                    | Retransmission and link instability RCA.                    |
| lte_rank          | MIMO rank or RI.                     | Spatial layer limitation RCA.                               |
| lte_rb_count      | Resource block allocation.           | Scheduler or resource limitation RCA.                       |
| carrier_count     | Number of LTE carriers.              | Carrier aggregation analysis.                               |

**Table 2:** Recommended canonical feature names.

## 4 Columns to Keep for Throughput Anomaly Detection

### 4.1 Feature Families

For the anomaly detection module, keep only columns that are directly related to throughput, explain throughput degradation, or help locate and report the anomaly.

| Feature Family      | Keep                                                                                        | Why It Matters                                                                                  |
|---------------------|---------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Metadata            | Session id, timestamp, route label, test type, session elapsed time, sample gap.            | Needed for grouping, time-series logic, train/test split, and traceability.                     |
| Location            | Latitude, longitude, speed, accuracy, altitude, distance from previous point.               | Needed for maps, speed effects, route clusters, and field engineer interpretation.              |
| LTE identity        | PCI, EARFCN, band, cell identity if available.                                              | Needed to group anomalies by serving cell, frequency, band, and route segment.                  |
| Radio quality       | RSRP, RSRQ, SINR, RSSI if available.                                                        | Main coverage and quality evidence for radio-limited degradation.                               |
| PHY and scheduler   | CQI, MCS, BLER, rank, resource blocks, modulation, bandwidth.                               | Explains why throughput is low even when coverage appears acceptable.                           |
| Carrier aggregation | Carrier count, SCell1 RSRP, SCell1 RSRQ, SCell1 SINR, SCell1 CQI, SCell1 BLER.              | Explains CA underperformance and secondary-cell limitations.                                    |
| Neighbor aggregates | Best neighbor RSRP, serving-minus-best-neighbor gap, neighbor count, strong-neighbor count. | Supports mobility and overlap/interference analysis without keeping many wide neighbor columns. |
| Event aggregates    | Event count near sample, handover-near flag, RACH-near flag, CA-near flag.                  | Adds RCA context from the event timeline.                                                       |

**Table 3:** Column families recommended for the LTE throughput anomaly module.

## 4.2 Leakage Columns to Exclude from Model Inputs

Some columns are useful for RCA but should not be used as predictors in the expected-throughput model because they directly contain the target or a direct component of the target.

### Avoid Target Leakage

Do not use total throughput, duplicate throughput columns, PCell throughput, SCell throughput, HTTP throughput, physical throughput, or activity throughput as model inputs when predicting LTE total downlink throughput. They can be used after anomaly detection for RCA comparison, but not as predictors.

Exclude from ML input features:

```

- LTE.LTE_Data_KPI.Agg_Throughput_DL # target
- LTE.DownlinkMeasurements.Throughput_DL # duplicate of target
- LTE - PrimaryCell - PCell_Throughput # direct part of target
- LTE - SecondaryCell 1 - SCell1_Throughput # direct part of target
- Throughput.HTTP_DL # application-layer target/reference
- Throughput.Physical_DL # physical/application reference
- Activity.Throughput # activity-level throughput

```

## 5 Throughput Anomaly Detection Strategy

### 5.1 Do Not Use One Generic Anomaly Label

A useful telecom anomaly pipeline should not only say *anomaly* or *normal*. It should classify what type of throughput degradation happened. This is important because a low-throughput sample caused by poor RSRP is different from a low-throughput sample that occurs despite excellent SINR and carrier aggregation.

| Anomaly Type                 | Detection Idea                                                               | Initial RCA Meaning                                                                                                           |
|------------------------------|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Radio-limited degradation    | Throughput is low and at least one of RSRP, RSRQ, SINR, or BLER is bad.      | Bad throughput is likely caused by weak coverage, poor quality, interference, or retransmissions.                             |
| Unexpected under-performance | Expected throughput is high, but actual throughput is much lower.            | Radio alone does not explain the degradation; investigate scheduler, CA, congestion, mobility, or server/application factors. |
| Sudden drop                  | Throughput drops sharply compared with previous samples in the same session. | Investigate whether the drop is synchronized with radio degradation or network events.                                        |
| CA underperformance          | Carrier count is high or CA is active, but throughput remains low.           | Investigate SCell weakness, low CA utilization, CA events, low rank, or low RB allocation.                                    |
| Event-related degradation    | Degradation appears near HO, RACH, CA, or other event windows.               | Mobility/access procedures may have disturbed the data session.                                                               |

**Table 4:** Recommended anomaly taxonomy.

### 5.2 Layer 1: Detect Throughput Degradation

The first layer detects whether throughput is poor. This can be done using fixed thresholds, session percentiles, or group percentiles.

```
df_lte["low_throughput_flag"] = df_lte["lte_dl_throughput"] < 10
```

A fixed threshold such as 10 Mbps is simple and easy to explain, but later the project should also include percentile-based thresholds because acceptable throughput depends on band, carrier count, radio quality, and test conditions.

### 5.3 Layer 2: Classify the Degradation Type

After detecting low throughput, classify the likely anomaly family. For example:

```
df_lte["poor_rsrp_flag"] = df_lte["lte_rsrp"] < -100
df_lte["poor_rsrq_flag"] = df_lte["lte_rsrq"] < -15
df_lte["poor_sinr_flag"] = df_lte["lte_sinr"] < 5

df_lte["radio_limited_degradation_flag"] = (
    df_lte["low_throughput_flag"] &
    (
        df_lte["poor_rsrp_flag"] |
        df_lte["poor_rsrq_flag"] |
        df_lte["poor_sinr_flag"]
    )
)
```

This keeps bad-radio cases inside the anomaly system instead of excluding them. The difference is that these cases are labeled as **radio-limited degradation**, not as unexplained underperformance.

## 6 Expected Throughput

### 6.1 Concept

Expected throughput means the throughput that should normally or reasonably be achieved under similar conditions. It is used to detect cases where the actual throughput is much lower than what the radio and PHY conditions suggest.

#### Important Distinction

Low throughput with poor RSRP or poor SINR is still an important anomaly, but it is usually radio-limited. The expected-throughput model is mainly used to find the other case: low throughput that is not fully explained by obvious bad radio conditions.

### 6.2 Method 1: Group-Based P75 Expected Throughput

The group-based method is the easiest baseline. It groups samples by similar conditions and calculates the 75th percentile of throughput inside each group.

```
df_lte["sinr_bin"] = pd.cut(
    df_lte["lte_sinr"],
```

```

    bins=[-50, 0, 13, 20, 50],
    labels=["Poor", "Fair", "Good", "Excellent"]
)

df_lte["rsrp_bin"] = pd.cut(
    df_lte["lte_rsrp"],
    bins=[-150, -100, -90, -80, 0],
    labels=["Poor", "Fair", "Good", "Excellent"]
)

group_cols = ["sinr_bin", "rsrp_bin", "carrier_count", "lte_band"]

expected_table = (
    df_lte.groupby(group_cols)["lte_dl_throughput"]
        .quantile(0.75)
        .reset_index()
        .rename(columns={"lte_dl_throughput": "expected_tp_p75"})
)

df_lte = df_lte.merge(expected_table, on=group_cols, how="left")

```

The 75th percentile represents a realistic achievable throughput. It is higher than the average but less extreme than the maximum, so it works well for optimization-oriented anomaly detection.

### 6.3 Group Size and Fallback

P75 only works well when the group has enough samples. Small groups should fall back to broader groups.

| Fallback Level | Grouping                                 | Use                                          |
|----------------|------------------------------------------|----------------------------------------------|
| Level 1        | SINR bin, RSRP bin, carrier count, band. | Most specific expected-throughput reference. |
| Level 2        | SINR bin, RSRP bin, carrier count.       | Used when band-level group is too small.     |
| Level 3        | SINR bin and carrier count.              | Broader fallback for sparse groups.          |
| Level 4        | Carrier count only.                      | Very broad fallback.                         |
| Level 5        | Global LTE-DL P75.                       | Last fallback.                               |

**Table 5:** Recommended fallback hierarchy for group-based expected throughput.

## 7 ML Regression Model for Expected Throughput

## 7.1 Recommended First Model

The first model should be a gradient-boosting tree model. The best practical starting point is:

- **HistGradientBoostingRegressor:** fast and strong for tabular data.
- **GradientBoostingRegressor with quantile loss:** useful later for predicting P75 achievable throughput.

### Recommended Modeling Sequence

Start with HistGradientBoostingRegressor to build a stable expected-throughput model. After the pipeline works, add a P75 quantile GradientBoostingRegressor to estimate achievable throughput.

## 7.2 How Gradient Boosting Works

Gradient boosting builds many small decision trees sequentially. Each new tree tries to correct the remaining error from the previous trees. For throughput, the model learns relationships such as:

- high SINR plus high carrier count usually increases expected throughput;
- poor RSRP, high BLER, or low CQI usually decreases expected throughput;
- SCell quality matters mainly when carrier aggregation is active;
- RB allocation and rank can explain low throughput even when RSRP looks acceptable.

This is better than a simple linear model because throughput depends on non-linear interactions between radio quality, scheduling, link adaptation, carrier aggregation, and mobility.

## 7.3 Session-Based Train/Test Split

Rows inside the same session are highly correlated. Therefore, the split must be based on session id, not random rows.

```
from sklearn.model_selection import GroupShuffleSplit

X = df_lte[feature_cols]
y = np.log1p(df_lte["lte_dl_throughput"])
groups = df_lte["id"]

splitter = GroupShuffleSplit(n_splits=1, test_size=0.2, random_state=42)
train_idx, test_idx = next(splitter.split(X, y, groups=groups))

X_train, X_test = X.iloc[train_idx], X.iloc[test_idx]
y_train, y_test = y.iloc[train_idx], y.iloc[test_idx]
```

## 7.4 Preprocessing and Training

```

from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import OneHotEncoder
from sklearn.impute import SimpleImputer
from sklearn.ensemble import HistGradientBoostingRegressor

numeric_features = X_train.select_dtypes(include=["int64", "float64", "bool"]).
    columns.tolist()
categorical_features = [c for c in X_train.columns if c not in numeric_features]

numeric_transformer = Pipeline(steps=[
    ("imputer", SimpleImputer(strategy="median"))
])

categorical_transformer = Pipeline(steps=[
    ("imputer", SimpleImputer(strategy="most_frequent")),
    ("onehot", OneHotEncoder(handle_unknown="ignore"))
])

preprocessor = ColumnTransformer(
    transformers=[
        ("num", numeric_transformer, numeric_features),
        ("cat", categorical_transformer, categorical_features),
    ]
)

model = HistGradientBoostingRegressor(
    max_iter=300,
    learning_rate=0.05,
    max_leaf_nodes=31,
    random_state=42
)

pipeline = Pipeline(steps=[
    ("preprocess", preprocessor),
    ("model", model)
])

pipeline.fit(X_train, y_train)

```

## 7.5 Evaluation Metrics

Evaluate the model on the test sessions using:

- **MAE:** average prediction error in Mbps; easy to explain.
- **RMSE:** penalizes large errors more strongly.
- **R2:** shows how much variation is explained by the model.
- **Residual plots:** show whether the model systematically fails in certain condi-

tions.

```
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

y_pred_log = pipeline.predict(X_test)
y_pred = np.expml(y_pred_log)
y_true = np.expml(y_test)

mae = mean_absolute_error(y_true, y_pred)
rmse = mean_squared_error(y_true, y_pred, squared=False)
r2 = r2_score(y_true, y_pred)
```

## 8 Anomaly Scores from Expected Throughput

After predicting expected throughput, calculate the gap and ratio.

```
df_lte["expected_tp_ml"] = np.expml(final_pipeline.predict(X))

df_lte["tp_gap"] = df_lte["expected_tp_ml"] - df_lte["lte_dl_throughput"]

df_lte["tp_ratio"] = (
    df_lte["lte_dl_throughput"] /
    df_lte["expected_tp_ml"].replace(0, np.nan)
)

df_lte["tp_log_residual"] = (
    np.log1p(df_lte["lte_dl_throughput"]) -
    np.log1p(df_lte["expected_tp_ml"])
)
```

| Score           | Meaning                                           | Anomaly Interpretation                                                            |
|-----------------|---------------------------------------------------|-----------------------------------------------------------------------------------|
| tp_gap          | Expected throughput minus actual throughput.      | Large positive gap means actual throughput is much lower than expected.           |
| tp_ratio        | Actual throughput divided by expected throughput. | Low ratio means the sample achieved only a small fraction of expected throughput. |
| tp_log_residual | Log-scale difference between actual and expected. | Useful because throughput is usually skewed.                                      |

**Table 6:** Main expected-throughput anomaly scores.

A first unexpected-underperformance rule can be:

```
df_lte["unexpected_underperformance_flag"] = (
    (df_lte["expected_tp_ml"] >= 20) &
    (df_lte["tp_ratio"] <= 0.40) &
```

```
(df_lte["tp_gap"] >= 10)
)
```

This means the model expected at least 20 Mbps, but the actual throughput was less than 40 percent of expected and the gap was at least 10 Mbps.

## 9 RCA of Throughput Anomalies

### 9.1 RCA Evidence Flags

The anomaly detection module should prepare RCA-ready flags even if another team/-module performs the final RCA decision. These flags convert raw KPIs into interpretable evidence.

| RCA Flag          | Condition Example                                | Interpretation                                              |
|-------------------|--------------------------------------------------|-------------------------------------------------------------|
| Poor RSRP         | RSRP below -100 dBm.                             | Weak coverage or cell-edge condition.                       |
| Poor RSRQ         | RSRQ below -15 dB.                               | Poor quality, interference, or loading issue.               |
| Poor SINR         | SINR below 5 dB.                                 | Interference or poor radio quality.                         |
| High BLER         | BLER above an agreed threshold.                  | Retransmissions reduce effective throughput.                |
| Low CQI           | CQI is low compared with expected radio quality. | UE reports poor channel quality.                            |
| Low MCS           | MCS is low.                                      | Link adaptation selected a conservative coding/modulation.  |
| Low RB allocation | Resource block allocation is low.                | Possible scheduler/resource limitation or congestion.       |
| Low rank          | Rank is low despite good radio.                  | Limited MIMO spatial layers.                                |
| Weak SCell        | SCell RSRP/SINR is poor.                         | CA may be active but the secondary cell contributes poorly. |
| Near HO event     | Sample is close to handover event.               | Throughput may be affected by mobility procedure.           |
| Near RACH failure | Sample is close to RACH failure.                 | Access procedure issue may affect data continuity.          |
| Near CA change    | Sample is close to CA activation/deactivation.   | CA state change may explain throughput instability.         |

**Table 7:** RCA evidence flags for throughput degradation.

## 9.2 RCA Decision Logic

The first RCA logic can be rule-based. It does not need to be perfect in V1; it should produce clear candidate causes and evidence.

```
def assign_primary_rca(row):
    if row["poor_rsrp_flag"]:
        return "Coverage-limited throughput degradation"
    if row["poor_sinr_flag"] or row["poor_rsrq_flag"]:
        return "Quality/interference-limited throughput degradation"
    if row.get("high_bler_flag", False):
        return "High BLER / retransmission-related degradation"
    if row.get("low_rb_allocation_flag", False):
        return "Scheduler or resource-allocation limitation"
```

```

if row.get("ca_underperformance_flag", False):
    return "Carrier aggregation underperformance"
if row.get("near_handover_flag", False):
    return "Mobility-related degradation near handover"
if row.get("near_rach_failure_flag", False):
    return "Access-related degradation near RACH issue"
if row["unexpected_underperformance_flag"]:
    return "Unexpected underperformance; hidden network or application cause"
return "No dominant RCA evidence"

```

### 9.3 Event-Window RCA

The `events_long.csv` table should be used to create event-window features. Instead of using raw event text in ML, create structured flags and counts.

Examples:

- number of handover events within  $\pm 5$  seconds;
- number of RACH failures within  $\pm 10$  seconds;
- number of CA activation/deactivation events within  $\pm 10$  seconds;
- time since last handover;
- time since last RACH failure;
- event density in the previous 30 seconds.

These features should be added after the first pipeline works. They will make the RCA much stronger because they connect low throughput with actual network procedures.

## 10 Final RCA-Ready Output Shape

The anomaly detection output should be designed for the RCA and LLM modules. Each anomaly row should contain location, session, actual/expected throughput, anomaly score, severity, and evidence flags.

Recommended anomaly output columns:

```

- anomaly_id
- id
- timestamp
- latitude
- longitude
- route_label
- test_type
- serving_pci
- serving_earfcn
- serving_band
- actual_lte_dl_throughput
- expected_lte_dl_throughput
- throughput_gap

```

```

- throughput_ratio
- anomaly_type
- anomaly_severity
- lte_rsrp
- lte_rsrq
- lte_sinr
- lte_cqi
- lte_mcs
- lte_bler
- lte_rank
- lte_rb_count
- carrier_count
- poor_rsrp_flag
- poor_rsrq_flag
- poor_sinr_flag
- high_bler_flag
- low_cqi_flag
- low_rb_allocation_flag
- ca_underperformance_flag
- near_handover_flag
- near_rach_failure_flag
- near_ca_change_flag
- primary_rca_candidate
- supporting_evidence_text

```

### LLM Integration

The LLM should not receive raw 1365-column data. It should receive the anomaly output and RCA evidence summary. The LLM task is to convert structured evidence into a professional explanation, not to discover the anomaly from raw samples.

## 11 Implementation Roadmap

### 11.1 Milestone 1: LTE-DL Modeling Table

1. Load the cleaned base parquet file.
2. Filter to LTE-DL rows only.
3. Rename important columns to canonical names.
4. Remove leakage columns from model inputs.
5. Export `lte_dl_modeling_table.parquet`.

### 11.2 Milestone 2: Baseline Anomaly Detection

1. Create simple low-throughput flags.
2. Create radio-limited degradation flags using RSRP, RSRQ, SINR, and BLER.

3. Create group-based P75 expected throughput.
4. Detect unexpected underperformance using expected throughput gap and ratio.
5. Export `lte_throughput_anomalies_baseline.csv`.

### 11.3 Milestone 3: ML Expected Throughput

1. Train HistGradientBoostingRegressor using session-based split.
2. Evaluate MAE, RMSE, R2, and residual plots.
3. Predict expected throughput for all LTE-DL rows.
4. Create ML residual anomaly score.
5. Export `lte_throughput_anomalies_ml.csv`.

### 11.4 Milestone 4: RCA Features

1. Add structured event-window features from `events_long.csv`.
2. Add CA, RACH, and handover proximity flags.
3. Add RCA reason flags and primary RCA candidate.
4. Export `lte_throughput_anomalies_rca_ready.csv`.

### 11.5 Milestone 5: LLM-Ready JSON

1. Convert each anomaly row into a compact evidence object.
2. Include only relevant KPIs, anomaly score, RCA flags, and event context.
3. Produce a consistent output schema for the LLM module.

## 12 Recommended First-Version Outputs

| Output                  | Purpose                                                  | User                                        |
|-------------------------|----------------------------------------------------------|---------------------------------------------|
| LTE-DL modeling table   | Clean LTE-DL table with target, features, and meta-data. | Anomaly detection module.                   |
| Group P75 reference     | Group-based expected throughput reference.               | Baseline anomaly detection and explanation. |
| Model evaluation report | Model metrics by train/test sessions.                    | Model validation.                           |
| Baseline anomaly table  | Rule-based and group-based anomalies.                    | First anomaly review.                       |
| ML anomaly table        | ML residual anomalies.                                   | Advanced anomaly detection.                 |
| RCA-ready anomaly table | Final anomaly table with RCA flags and evidence.         | RCA and LLM modules.                        |
| Anomaly route map       | Map of anomaly locations.                                | Visual analysis and reporting.              |

**Table 8:** Recommended first-version outputs.

```
Suggested filenames:
- lte_dl_modeling_table.parquet
- expected_throughput_group_p75.csv
- model_evaluation_report.csv
- lte_throughput_anomalies_baseline.csv
- lte_throughput_anomalies_ml.csv
- lte_throughput_anomalies_rca_ready.csv
- throughput_anomaly_map.html
```

## 13 Columns to Bring Back Later

The first version can work without restoring removed columns. However, once the pipeline is stable, several sparse but important columns should be brought back because they improve throughput RCA.

```
High-priority columns to restore later:

Application-layer throughput:
- Throughput.HTTP_DL
- Throughput.Physical_DL

Handover interruption:
- Extra_KPIs.LTE Handover Interruption Time
```

```
- Extra_KPIs.UMTS Handover Interruption Time

Call/accessibility timing, useful for later MO/CSFB modules:
- Extra_KPIs.ERRC Connection Setup Time
- Extra_KPIs.RRC Connection Setup Time
- Extra_KPIs.MO Call Setup Time
- Extra_KPIs.MO Call Setup Time Until Connect
- Extra_KPIs.MO CSFB Call Setup Time
- Extra_KPIs.Attach Delay

Activity/application RCA:
- Activity.Error
- Activity.TTFB
- Activity.DNS Resolution Time
- Activity.Remote Address
- Activity.Remote IPs
- Activity.Size
- Activity.ARFCN
- Activity.Band
- Activity.LCI
```

### 13.1 Better Than Restoring All Neighbor Columns

Instead of restoring many wide neighbor columns, create long-format neighbor tables:

- `lte_neighbors_long.csv`: one row per LTE neighbor observation.
- `wcdma_cells_long.csv`: one row per WCDMA detected/active-set cell.

This approach keeps the main modeling table clean while still preserving mobility and neighbor information for RCA.

## 14 Quality Checks and Risks

### Main Risks

The three biggest risks are target leakage, mixing different service types, and over-trusting row-random train/test splits. These must be controlled from the first notebook version.

| Risk                    | Problem                                                                     | Control                                                                        |
|-------------------------|-----------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Target leakage          | The model may use throughput-derived columns to predict throughput.         | Exclude PCell/SCell throughput and duplicate throughput columns from features. |
| Mixed service types     | LTE-DL, MO call, and WCDMA behavior have different meanings.                | Start with LTE-DL rows only; build separate modules later.                     |
| Random split leakage    | Rows in the same session are similar, causing unrealistic test performance. | Split by session id using GroupShuffleSplit.                                   |
| Small P75 groups        | Group-based expected throughput becomes unstable.                           | Add group count and fallback hierarchy.                                        |
| Over-flagging bad radio | All bad-radio low-throughput samples may be marked as unexpected anomalies. | Separate radio-limited degradation from unexpected underperformance.           |
| Raw event text noise    | Event descriptions are messy and hard to model directly.                    | Convert events into structured event-window flags.                             |

**Table 9:** Main risks and controls.

## 15 Conclusion

The chosen direction is technically correct. The strongest first module is LTE downlink throughput anomaly detection, built on cleaned LTE-DL samples only. The anomaly system should not only detect low throughput; it should classify degradation into radio-limited cases, unexpected underperformance, CA underperformance, event-related cases, and sudden drops.

The expected-throughput approach is central because it separates throughput that is low for obvious radio reasons from throughput that is unexpectedly low despite favorable conditions. For the first version, implement a group-based P75 baseline and a HistGradientBoostingRegressor model. After that, add quantile regression, event-window features, handover/RACH/CA flags, and restored sparse throughput/RCA columns.

The final output should be RCA-ready and LLM-ready: each anomaly should include actual throughput, expected throughput, anomaly severity, location, serving cell context, radio evidence, PHY evidence, event evidence, and a primary RCA candidate.